

Electrochemistry

Chapter 2 · All questions with full solutions



Part 1 · Examples

Q

2.1

Cell notation
and Nernst
equation



Represent the cell in which the following reaction takes place $\text{Mg(s)} + 2 \text{Ag}^+ (0.0001 \text{ M}) \rightarrow \text{Mg}^{2+} (0.130 \text{ M}) + 2 \text{Ag(s)}$
Calculate its $E_{(\text{cell})}$ if $E_{(\text{cell})}^{\circ} = 3.17 \text{ V}$.

Given

The cell can be written as $\text{Mg} | \text{Mg}^{2+} (0.130 \text{ M}) || \text{Ag}^+ (0.0001 \text{ M}) | \text{Ag}$

Key formula



$$E_{(\text{cell})} = E_{(\text{cell})}^{\circ} - \frac{RT}{2F} \ln \frac{\text{Mg}^{2+}}{\text{Ag}^{+2}}$$

$$= 3.17 \text{ V} - \frac{0.059 \text{ V}}{2} \log \frac{0.130}{(0.0001)^2} = 3.17 \text{ V} - 0.21 \text{ V} = 2.96 \text{ V}$$

$$E_{(\text{cell})} = 2.96 \text{ V} //$$

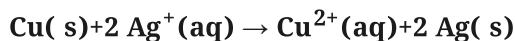
Q

2.2

Equilibrium
constant
from
standard cell
potential



Calculate the equilibrium constant of the reaction:



$$E_{(\text{cell})}^{\circ} = 0.46 \text{ V}$$

Key formula



$$E_{(\text{cell})}^{\circ} = \frac{0.059 \text{ V}}{2} \log K_C = 0.46 \text{ V or}$$

$$\log K_C = \frac{0.46 \text{ V} \times 2}{0.059 \text{ V}} = 15.6$$

$$K_C = 3.92 \times 10^{15}$$

$$K_C = 3.92 \times 10^{15} //$$

Q

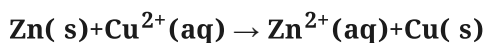
2.3

Standard
Gibbs energy
from cell
potential

((

The standard electrode potential for Daniell cell is 1.1V.

Calculate the standard Gibbs energy for the reaction:



Key formula



$$\Delta_r G^{\circ} = -n F E_{(\text{cell})}^{\circ}$$

Given

n in the above equation is 2, $F = 96487 \text{ C mol}^{-1}$ and $E_{(\text{cell})}^{\circ} = 1.1 \text{ V}$

Conclusion

Therefore,

$$\Delta_r G^0 = -2 \times 1.1 \text{ V} \times 96487 \text{ C mol}^{-1}$$

$$= -21227 \text{ J mol}^{-1}$$

$$= -212.27 \text{ kJ mol}^{-1}$$

$$\Delta_r G^0 = -212.27 \text{ kJ mol}^{-1} //$$

Q**2.4**

Conductivity
and molar
conductivity
from cell
constant



Resistance of a conductivity cell filled with 0.1 mol L^{-1} KCl solution is 100Ω . If the resistance of the same cell when filled with 0.02 mol L^{-1} KCl solution is 520Ω , calculate the conductivity and molar conductivity of 0.02 mol L^{-1} KCl solution. The conductivity of 0.1 mol L^{-1} KCl solution is 1.29 S / m .

Given

The cell constant is given by the equation:

**Key formula**Cell constant = G^* = conductivity $\times$ resistance**Substitute**

$$= 1.29 \text{ S / m} \times 100 \Omega = 129 \text{ m}^{-1} = 1.29 \text{ cm}^{-1}$$

Key formulaConductivity of 0.02 mol L^{-1} KCl solution = cell constant / resistance**Substitute**

$$= \frac{G^*}{R} = \frac{129 \text{ m}^{-1}}{520 \Omega} = 0.248 \text{ S m}^{-1}$$

Concentration

$$= 0.02 \text{ mol L}^{-1}$$

$$= 1000 \times 0.02 \text{ mol m}^{-3} = 20 \text{ mol m}^{-3}$$

Key formulaMolar conductivity = $\Lambda_m = \frac{\kappa}{c}$ **Conclusion**

$$= \frac{248 \times 10^{-3} \text{ S m}^{-1}}{20 \text{ mol m}^{-3}} = 124 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$$

Conductivity = $0.248 \times 10^{-2} \text{ S cm}^{-1}$ (= 0.248 S m^{-1}); **Molar conductivity** $\Lambda_m = 124 \text{ S cm}^2 \text{ mol}^{-1}$

//

Alternative method Alternatively,

$$\kappa = \frac{1.29 \text{ cm}^{-1}}{520 \Omega} = 0.248 \times 10^{-2} \text{ S cm}^{-1}$$

and

$$\Lambda_m = \kappa \times 1000 \text{ cm}^3 \text{ L}^{-1} \text{ molarity}^{-1}$$

$$= \frac{0.248 \times 10^{-2} \text{ S cm}^{-1} \times 1000 \text{ cm}^3 \text{ L}^{-1}}{0.02 \text{ mol L}^{-1}}$$

$$= 124 \text{ S cm}^2 \text{ mol}^{-1}$$

Q

2.5

Resistivity, conductivity and molar conductivity of NaOH



The electrical resistance of a column of 0.05 mol L^{-1} NaOH solution of diameter 1 cm and length 50 cm is $5.55 \times 10^3 \text{ ohm}$. Calculate its resistivity, conductivity and molar conductivity.

Given

$$A = \pi r^2 = 3.14 \times 0.5^2 \text{ cm}^2 = 0.785 \text{ cm}^2 = 0.785 \times 10^{-4} \text{ m}^2 \quad l = 50 \text{ cm} = 0.5 \text{ m}$$

Substitute

$$R = \frac{\rho l}{A} \quad \text{or} \quad \rho = \frac{R A}{l} = \frac{5.55 \times 10^3 \Omega \times 0.785 \text{ cm}^2}{50 \text{ cm}} = 87.135 \Omega \text{ cm}$$

$$\text{Conductivity} = \kappa = \frac{1}{\rho} = \left(\frac{1}{87.135} \right) \text{ S cm}^{-1}$$

$$= 0.01148 \text{ S cm}^{-1}$$

Key formula

$$\text{Molar conductivity, } \Lambda_m = \frac{\kappa \times 1000}{c} \text{ cm}^3 \text{ L}^{-1}$$



Conclusion

$$= \frac{0.01148 \text{ S cm}^{-1} \times 1000 \text{ cm}^3 \text{ L}^{-1}}{0.05 \text{ mol L}^{-1}}$$

$$= 229.6 \text{ S cm}^2 \text{ mol}^{-1}$$

Resistivity $\rho = 87.135 \Omega \text{ cm}$; Conductivity $\kappa = 0.01148 \text{ S cm}^{-1}$; Molar conductivity $\Lambda_m = 229.6 \text{ S cm}^2 \text{ mol}^{-1}$



Alternative units

If we want to calculate the values of different quantities in terms of 'm' instead of 'cm',

$$\rho = \frac{R A}{l}$$

$$= \frac{5.55 \times 10^3 \Omega \times 0.785 \times 10^{-4} \text{ m}^2}{0.5 \text{ m}} = 87.135 \times 10^{-2} \Omega \text{ m}$$

$$\kappa = \frac{1}{\rho} = \frac{100}{87.135} \Omega \text{ m} = 1.148 \text{ S m}^{-1}$$

$$\text{and } \Lambda_m = \frac{\kappa}{c} = \frac{1.148 \text{ S m}^{-1}}{50 \text{ mol m}^{-3}} = 229.6 \times 10^{-4} \text{ S m}^2 \text{ mol}^{-1}$$

Q

2.6

Limiting molar conductivity of KCl by extrapolation



The molar conductivity of KCl solutions at different concentrations at 298 K are given below:

$c / \text{mol l}^{-1}$	$\Lambda_m / \text{S cm}^2 \text{mol}^{-1}$
0.000198	148.61
0.000309	148.29
0.000521	147.81
0.000989	147.09

Show that a plot between Λ_m and $c^{1/2}$ is a straight line. Determine the values of Λ_m° and A for KCl.

Given

Taking the square root of concentration, we get:

$$c^{1/2} / (\text{mol l}^{-1})^{1/2} \quad \Lambda_m / \text{S cm}^2 \text{mol}^{-1}$$

$$0.01407 \quad 148.61$$

$$0.01758 \quad 148.29$$

$$0.02283 \quad 147.81$$

$$0.03145 \quad 147.09$$

A plot of Λ_m (y-axis) against $c^{1/2}$ (x-axis) is shown in (Fig. 3.7). It is nearly a straight line. From the intercept ($c^{1/2} = 0$), we get:

Conclusion

$$\Lambda_m^\circ = 150.0 \text{ S cm}^2 \text{mol}^{-1} \text{ and}$$

$$A = -\text{slope} = 87.46 \text{ S cm}^2 \text{mol}^{-1} / (\text{mol l}^{-1})^{1/2}$$

$$\Lambda_m^\circ = 150.0 \text{ S cm}^2 \text{mol}^{-1}; A = 87.46 \text{ S cm}^2 \text{mol}^{-1} / (\text{mol l}^{-1})^{1/2}$$

///

Q

2.7

Limiting molar conductivity via Kohlrausch's law

()

Calculate Λ_m° for CaCl_2 and MgSO_4 from the data given in Table 3.4.

Given

We know from Kohlrausch law that

Conclusion

$$\Lambda_m(\text{CaCl}_2)^\circ = \lambda_{\text{Ca}^{2+}}^\circ + 2\lambda_{\text{Cl}^-}^\circ = 119.0 \text{ S cm}^2 \text{mol}^{-1} + 2(76.3) \text{ S cm}^2 \text{mol}^{-1}$$

$$= (119.0 + 152.6) \text{ S cm}^2 \text{mol}^{-1}$$

$$= 271.6 \text{ S cm}^2 \text{mol}^{-1}$$

$$\Lambda_m(\text{MgSO}_4)^\circ = \lambda_{\text{Mg}^{2+}}^\circ + \lambda_{\text{SO}_4^{2-}}^\circ = 106.0 \text{ S cm}^2 \text{mol}^{-1} + 160.0 \text{ S cm}^2 \text{mol}^{-1}$$

$$= 266 \text{ S cm}^2 \text{mol}^{-1}$$

$$\Lambda_m^\circ(\text{CaCl}_2) = 271.6 \text{ S cm}^2 \text{mol}^{-1}; \Lambda_m^\circ(\text{MgSO}_4) = 266 \text{ S cm}^2 \text{mol}^{-1}$$

///

Q

2.8

Limiting molar conductivity of a weak electrolyte via Kohlrausch's law



Λ_m^0 for NaCl, HCl and NaAc are 126.4, 425.9 and 91.0 S cm² mol⁻¹ respectively.

Calculate Λ^0 for HAc.

Key formula

$$\Lambda_{m(\text{HAc})}^0 = \lambda_{\text{H}^+}^0 + \lambda_{\text{Ac}^-}^0 = \lambda_{\text{H}^+}^0 + \lambda_{\text{Cl}^-}^0 + \lambda_{\text{Ac}^-}^0 + \lambda_{\text{Na}^+}^0 - \lambda_{\text{Cl}^-}^0 - \lambda_{\text{Na}^+}^0$$



Conclusion

$$= \Lambda_{m(\text{HCl})}^0 + \Lambda_{m(\text{NaAc})}^0 - \Lambda_{m(\text{NaCl})}^0$$

$$= (425.9 + 91.0 - 126.4) \text{ S cm}^2 \text{ mol}^{-1}$$

$$= 390.5 \text{ S cm}^2 \text{ mol}^{-1}$$

$$\Lambda_{m(\text{HAc})}^0 = 390.5 \text{ S cm}^2 \text{ mol}^{-1} //$$

Q

2.9

Dissociation constant of acetic acid from conductivity



The conductivity of 0.001028 mol L⁻¹ acetic acid is 4.95 × 10⁻⁵ S cm⁻¹. Calculate its dissociation constant if Λ_m^0 for acetic acid is 390.5 S cm² mol⁻¹.

Substitute

$$\Lambda_m = \frac{\kappa}{c} = \frac{4.95 \times 10^{-5} \text{ Scm}^{-1}}{0.001028 \text{ mol L}^{-1}} \times \frac{1000 \text{ cm}^3}{\text{L}} = 48.15 \text{ S cm}^2 \text{ mol}^{-1}$$

Conclusion

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0} = \frac{48.15 \text{ Scm}^2 \text{ mol}^{-1}}{390.5 \text{ Scm}^2 \text{ mol}^{-1}} = 0.1233$$

$$K_a = \frac{c \alpha^2}{(1-\alpha)} = \frac{0.001028 \text{ mol L}^{-1} \times (0.1233)^2}{1-0.1233} = 1.78 \times 10^{-5} \text{ mol L}^{-1}$$

$$\alpha = 0.1233; K_a = 1.78 \times 10^{-5} \text{ mol L}^{-1} //$$

Q

2.10

Mass of copper deposited via Faraday's law



A solution of CuSO₄ is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode?

Given

$$t = 600 \text{ s}$$

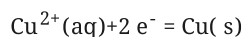


Substitute

$$\text{charge} = \text{current} \times \text{time} = 1.5 \text{ A} \times 600 \text{ s} = 900 \text{ C}$$

According to the reaction:

Key formula



Substitute

We require 2F or 2 × 96487 C to deposit 1 mol or 63 g of Cu .

For 900 C, the mass of Cu deposited

Conclusion

$$= (63 \text{ g mol}^{-1} \times 900 \text{ C}) / (2 \times 96487 \text{ C mol}^{-1}) = 0.2938 \text{ g} .$$

$$\text{Mass of Cu deposited} = 0.2938 \text{ g} //$$

Part 2 · Additional Questions

Q

1

Standard
electrode
potential of
Mg

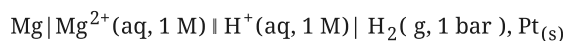
(())

How would you determine the standard electrode potential of the system $\text{Mg}^{2+} | \text{Mg}$?

Given

The standard electrode potential of $\text{Mg}^{2+} | \text{Mg}$ can be measured with respect to the standard hydrogen electrode, represented by $\text{Pt}_{(s)}, \text{H}_{2(g)} (1 \text{ atm}) | \text{H}^+_{(aq)} (1 \text{ M})$.

We set up a cell with $\text{Mg} | \text{MgSO}_4(aq, 1 \text{ M})$ as the anode and the standard hydrogen electrode as the cathode.



We then measure the emf of the cell.

This measured emf is the standard electrode potential of the magnesium electrode.

Key formula

$$E^\ominus = E_R^\ominus - E_L^\ominus$$



Substitute

Here, $E_R^\ominus$ for the standard hydrogen electrode is zero.

$$\therefore E^\ominus = 0 - E_L^\ominus$$

$$= -E_L^\ominus$$

Q

2

Zinc pot with
copper
sulphate



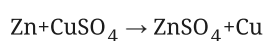
Can you store copper sulphate solutions in a zinc pot?

Conclusion

Zinc is more reactive than copper.

So zinc displaces copper from its salt solution.

If copper sulphate solution is stored in a zinc pot, zinc will displace the copper from it.



So, copper sulphate solution cannot be stored in a zinc pot.

Q

3

Oxidising
agents for
ferrous ions



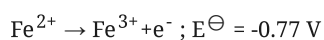
Consult the table of standard electrode potentials and suggest three substances that can oxidise ferrous ions under suitable conditions.

Given

Substances that are stronger oxidising agents than ferrous ions can oxidise ferrous ions.



Key formula



Conclusion

This implies that the substances having higher reduction potentials than $+0.77 \text{ V}$ can oxidise ferrous ions to ferric ions. Three substances that can do so are F_2 , Cl_2 , and O_2 .

Q

4

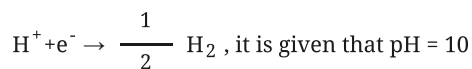
Hydrogen electrode potential at pH 10



Calculate the potential of hydrogen electrode in contact with a solution whose pH is 10.

Given

For hydrogen electrode,



$$\therefore [H^+] = 10^{-10} \text{ M}$$

Substitute

Now, using Nernst equation:

$$E_{(H^+, \frac{1}{2} H_2)} = E_{(H^+, \frac{1}{2} H_2)}^\ominus - \frac{RT}{nF} \ln \frac{1}{[H^+]}$$

$$= E_{(H^+, \frac{1}{2} H_2)}^\ominus - \frac{0.0591}{1} \log \frac{1}{[H^+]}$$

$$= 0 - \frac{0.0591}{1} \log \frac{1}{[10^{-10}]}$$

$$= -0.0591 \log 10^{10}$$

$$= -0.591 \text{ V}$$

-0.591 V //

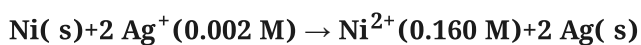
Q

5

EMF from Nernst equation

(())

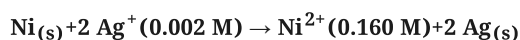
Calculate the emf of the cell in which the following reaction takes place:



Given that $E_{cell}^\ominus = 1.05 \text{ V}$

Conclusion

Applying Nernst equation we have:



$$E_{(cell)}^\ominus = 1.05 \text{ V}$$

$$E_{(cell)} = E_{(cell)}^\ominus - \frac{0.0591}{n} \log \frac{[Ni^{2+}]}{[Ag^+]^2}$$

$$= 1.05 - \frac{0.0591}{2} \log \frac{(0.160)}{(0.002)^2}$$

$$= 1.05 - 0.02955 \log \frac{0.16}{0.000004}$$

$$= 1.05 - 0.02955 \log 4 \times 10^4$$

$$= 1.05 - 0.02955(\log 10000 + \log 4)$$

$$= 1.05 - 0.02955(4 + 0.6021)$$

$$= 0.914 \text{ V}$$

0.914 V //

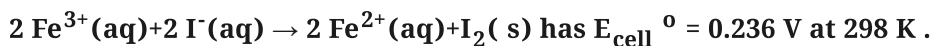
Q

6

Gibbs energy
and
equilibrium
constant



The cell in which the following reaction occurs:



Calculate the standard Gibbs energy and the equilibrium constant of the cell reaction.

Given

Here, $n = 2$, $E_{\text{cell}}^{\ominus} = 0.236 \text{ V}$, $T = 298 \text{ K}$



Substitute

We know that:

$$\begin{aligned} \Delta_r G^{\ominus} &= -n F E_{\text{cell}}^{\ominus} \\ &= -2 \times 96487 \times 0.236 \\ &= -45541.864 \text{ J mol}^{-1} \\ &= -45.54 \text{ kJ mol}^{-1} \end{aligned}$$

Key formula

Again, $\Delta_r G^{\ominus} = -2.303 R T \log K_c$



Conclusion

$$\begin{aligned} \Rightarrow \log K_c &= - \frac{\Delta_r G^{\ominus}}{2.303 R T} \\ &= - \frac{-45.54 \times 10^3}{2.303 \times 8.314 \times 298} \\ &= 7.981 \\ \therefore K_c &= \text{Antilog}(7.981) \\ &= 9.57 \times 10^7 \end{aligned}$$

$$\Delta_r G^{\ominus} = -45.54 \text{ kJ mol}^{-1}; K_c = 9.57 \times 10^7 //$$

Q

7

Conductivity
and dilution



Why does the conductivity of a solution decrease with dilution?

Conclusion

The conductivity of a solution is the conductance of ions present in a unit volume of the solution.

The number of ions (responsible for carrying current) decreases when the solution is diluted.

As a result, the conductivity of a solution decreases with dilution.

Q

8

Limiting
molar
conductivity
of water



Suggest a way to determine the Λ_m° value of water.

Given

Applying Kohlrausch's law of independent migration of ions, the Λ° value of water can be determined as follows:



Key formula

$$\begin{aligned}
 \Lambda_{m(\text{H}_2\text{O})}^0 &= \lambda_{\text{H}^+}^0 + \lambda_{\text{OH}^-}^0 \\
 &= (\lambda_{\text{H}^+}^0 + \lambda_{\text{Cl}^-}^0) + (\lambda_{\text{Na}^+}^0 + \lambda_{\text{OH}^-}^0) - (\lambda_{\text{Na}^+}^0 + \lambda_{\text{Cl}^-}^0) \\
 &= \Lambda_{m(\text{HCl})}^0 + \Lambda_{m(\text{NaOH})}^0 - \Lambda_{m(\text{NaCl})}^0 \\
 &= \Lambda_m^0
 \end{aligned}$$

Conclusion

Hence, by knowing the values of HCl, NaOH, and NaCl, the Λ_m^0 value of water can be determined.

Q**9**

Degree of dissociation of methanoic acid

(())

The molar conductivity of 0.025 mol L^{-1} methanoic acid is $46.1 \text{ S cm}^2 \text{ mol}^{-1}$. Calculate its degree of dissociation and dissociation constant. Given $\lambda^0(\text{H}^+) = 349.6 \text{ S cm}^2 \text{ mol}^{-1}$ and $\lambda^0(\text{HCOO}^-) = 54.6 \text{ S cm}^2 \text{ mol}^{-1}$.

Given

$$\begin{aligned}
 C &= 0.025 \text{ mol L}^{-1} \\
 \Lambda_m &= 46.1 \text{ S cm}^2 \text{ mol}^{-1} \\
 \lambda^0(\text{H}^+) &= 349.6 \text{ S cm}^2 \text{ mol}^{-1} \\
 \lambda^0(\text{HCOO}^-) &= 54.6 \text{ S cm}^2 \text{ mol}^{-1} \\
 \Lambda_m^0(\text{HCOOH}) &= \lambda^0(\text{H}^+) + \lambda^0(\text{HCOO}^-) \\
 &= 349.6 + 54.6 \\
 &= 404.2 \text{ S cm}^2 \text{ mol}^{-1}
 \end{aligned}$$

Substitute

Now, degree of dissociation:

$$\begin{aligned}
 \alpha &= \frac{\Lambda_m(\text{HCOOH})}{\Lambda_m^0(\text{HCOOH})} \\
 &= \frac{46.1}{404.2} \\
 &= 0.114 \text{ (approximately)}
 \end{aligned}$$

Conclusion

Thus, dissociation constant:

$$\begin{aligned}
 K &= \frac{c \alpha^2}{(1-\alpha)} \\
 &= \frac{(0.025 \text{ mol L}^{-1})(0.114)^2}{(1-0.114)} \\
 &= 3.67 \times 10^{-4} \text{ mol L}^{-1}
 \end{aligned}$$

$$\alpha = 0.114 \text{ (approximately); } K = 3.67 \times 10^{-4} \text{ mol L}^{-1}$$

///

Q

10

Electrons
from current
and time

If a current of 0.5 ampere flows through a metallic wire for 2 hours, then how many electrons would flow through the wire?

Given

The current is:

$$= 0.5 \text{ A}$$

$$t = 2 \text{ hours} = 2 \times 60 \times 60 \text{ s} = 7200 \text{ s}$$

Substitute

Thus, $Q = I t$

$$= 0.5 \text{ A} \times 7200 \text{ s}$$

$$= 3600 \text{ C}$$

We know that 96487 C = 6.023×10^{23} gives the number of electrons. So:

$$3600 \text{ C} = \frac{6.023 \times 10^{23} \times 3600}{96487} \text{ number of electrons}$$

$$= 2.25 \times 10^{22} \text{ number of electrons}$$

Conclusion

Hence, 2.25×10^{22} number of electrons will flow through the wire.

$$2.25 \times 10^{22} \text{ electrons} //$$

Q

11

Metals
extracted
electrolytical
ly

Suggest a list of metals that are extracted electrolytically.

Conclusion

Metals that are on the top of the reactivity series such as sodium, potassium, calcium, lithium, magnesium, aluminium are extracted electrolytically.

Sodium, potassium, calcium, lithium, magnesium, aluminium //

Q

12

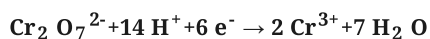
Charge to
reduce
dichromate

Consider the reaction: $\text{Cr}_2 \text{O}_7^{2-} + 14 \text{H}^+ + 6 \text{e}^- \rightarrow 2 \text{Cr}^{3+} + 7 \text{H}_2 \text{O}$

What is the quantity of electricity in coulombs needed to reduce 1 mol of $\text{Cr}_2 \text{O}_7^{2-}$?

Conclusion

The given reaction is as follows:



Therefore, to reduce 1 mole of $\text{Cr}_2 \text{O}_7^{2-}$, the required quantity of electricity will
= 6 F

$$= 6 \times 96487 \text{ C}$$

$$= 578922 \text{ C}$$

$$578922 \text{ C (6 F)} //$$

Q

13

Recharging
lead storage
battery

(())

Write the chemistry of recharging the lead storage battery, highlighting all the materials that are involved during recharging.

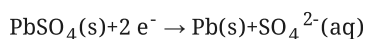
Not in official solutions manual: This answer was computed for completeness, since the official English solutions manual does not contain a solution for this question.

Given

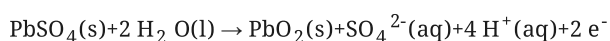
On recharging, an external source of current is applied to the lead storage battery and the spontaneous discharge reaction is reversed; the electrode that was the anode during discharge acts as the cathode during charging, and the electrode that was the cathode during discharge acts as the anode during charging.

Key formula

At the cathode (charging), lead sulphate is reduced back to spongy lead:



At the anode (charging), lead sulphate is oxidised back to lead dioxide:



Conclusion

Overall reaction on recharging (the exact reverse of the discharge reaction):



The materials involved are solid lead (Pb) and solid lead dioxide (PbO₂), both regenerated from the lead sulphate (PbSO₄) coating both electrodes after discharge, plus water and sulphuric acid in the electrolyte, whose concentration is restored as the two half-reactions proceed.

Q

14

Alternative
fuel cell
materials

Suggest two materials other than hydrogen that can be used as fuels in fuel cells.

Conclusion

Methane and methanol can be used as fuels in fuel cells.

Methane and methanol //

Q

15

Rusting as
electrochemi
cal cell

Explain how rusting of iron is envisaged as setting up of an electrochemical cell.

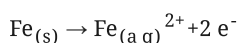
Given

In corrosion, air and moisture cause oxidation at one spot on an iron object.

This spot acts as the anode.

The anode reaction is:

Key formula



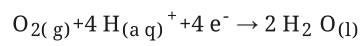
Substitute

Electrons released at this anodic spot travel through the metal to another spot on the same object.

At that spot, in the presence of H⁺ ions, the electrons reduce oxygen. This spot acts as the cathode. These H⁺ ions come either from H₂CO₃, formed when carbon dioxide from air dissolves in water; or from other acidic oxides in the atmosphere dissolving in water.

The cathode reaction is:

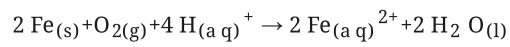
Key formula



Substitute

The overall reaction is:

Key formula



Conclusion

Atmospheric oxygen then oxidises the ferrous ions further to ferric ions. These ferric ions combine with moisture in the surroundings to form hydrated ferric oxide, $(\text{Fe}_2 \text{O}_3 \cdot x \text{H}_2 \text{O})_{\text{i.e., rust.}}$

So, rusting of iron can be seen as the setting up of an electrochemical cell.

Q

1

Metal
displacement
order



Arrange the following metals in the order in which they displace each other from the solution of their salts.

Al, Cu, Fe, Mg and Zn.

Conclusion

The following is the order in which the given metals displace each other from the solution of their salts.

Mg, Al, Zn, Fe, Cu

Mg, Al, Zn, Fe, Cu //

Q

2

Reducing
power
ranking

(())

Given the standard electrode potentials,

$K^+ / K = -2.93 \text{ V}$, $Ag^+ / Ag = 0.80 \text{ V}$,

$Hg^{2+} / Hg = 0.79 \text{ V}$

$Mg^{2+} / Mg = -2.37 \text{ V}$, $Cr^{3+} / Cr = -0.74 \text{ V}$

Arrange these metals in their increasing order of reducing power.

Conclusion

The lower the reduction potential, the higher is the reducing power. The given standard electrode potentials increase in the order of

$K^+ / K < Mg^{2+} / Mg < Cr^{3+} / Cr < Hg^{2+} / Hg < Ag^+ / Ag$.

Hence, the reducing power of the given metals increases in the following order:

$Ag < Hg < Cr < Mg < K$

$Ag < Hg < Cr < Mg < K //$

Q

3

Zn-Ag
galvanic cell
setup



Depict the galvanic cell in which the reaction

$Zn(s) + 2 Ag^+(aq) \rightarrow Zn^{2+}(aq) + 2 Ag(s)$ takes place. Further show:

Part (i)

Which of the electrode is negatively charged?

Part (ii)

The carriers of the current in the cell.

Part (iii)

Individual reaction at each electrode.

Conclusion

The galvanic cell in which the given reaction takes place is depicted as:

$Zn_{(s)} | Zn_{(aq)}^{2+} || Ag_{(aq)}^+ | Ag_{(s)}$

(i) Zn electrode (anode) is negatively charged.

(ii) Ions carry the current in the cell.

In the external circuit, current flows from silver to zinc.

(iii) The reaction taking place at the anode is given by,

$Zn_{(s)} \rightarrow Zn_{(aq)}^{2+} + 2 e^-$

The reaction taking place at the cathode is given by,

$Ag_{(aq)}^+ + e^- \rightarrow Ag_{(s)}$

Q

4

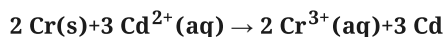
Standard cell potential and Gibbs energy



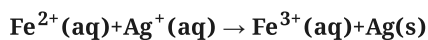
Calculate the standard cell potentials of galvanic cell in which the following reactions take place:

Calculate the $\Delta_r G^\ominus$ and equilibrium constant of the reactions.

Part (i)



Part (ii)



Given

$$(i) E_{\text{Cr}^{3+} / \text{Cr}}^\ominus = -0.74 \text{ V}$$

$$E_{\text{Cd}^{2+} / \text{Cd}}^\ominus = -0.40 \text{ V}$$

The galvanic cell of the given reaction is depicted as:



Substitute

Now, the standard cell potential is

$$E_{\text{cell}}^\ominus = E_{\text{R}}^\ominus - E_{\text{L}}^\ominus$$

$$= -0.40 - (-0.74)$$

$$= +0.34 \text{ V}$$

$$\Delta_r G^\ominus = -n F E_{\text{cell}}^\ominus$$

In the given equation, n

$$= 6$$

$$F = 96487 \text{ C mol}^{-1}$$

$$E_{\text{cell}}^\ominus = +0.34 \text{ V}$$

$$\text{Then, } \Delta_r G^\ominus = -6 \times 96487 \text{ C mol}^{-1} \times 0.34 \text{ V}$$

$$= -196833.48 \text{ CV mol}^{-1}$$

$$= -196833.48 \text{ J mol}^{-1}$$

$$= -196.83 \text{ kJ mol}^{-1}$$

Conclusion

Again,

$$\Delta_r G^\ominus = -R T \ln K$$

$$\Rightarrow \Delta_r G^\ominus = -2.303 R T \ln K$$

$$\Rightarrow \log K = - \frac{\Delta_r G}{2.303 R T}$$

$$= \frac{-196.83 \times 10^3}{2.303 \times 8.314 \times 298}$$

$$= 34.496$$

$$\therefore K = \text{antilog}(34.496) =$$

$$3.13 \times 10^{34}$$

Given

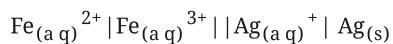
(ii)

$$E_{\text{Fe}^{3+} / \text{Fe}^{2+}} = 0.77 \text{ V}$$



$$E_{\text{Ag}^+ / \text{Ag}}^\ominus = 0.80 \text{ V}$$

The galvanic cell of the given reaction is depicted as:



Substitute

Now, the standard cell potential is

$$E_{\text{cell}}^\ominus = E_{\text{R}}^\ominus - E_{\text{L}}^\ominus$$

$$= 0.80 - 0.77$$

$$= 0.03 \text{ V}$$

Here, $n = 1$.

$$\text{Then, } \Delta_{\text{r}} G^\ominus = -n F E_{\text{cell}}^\ominus$$

$$= -1 \times 96487 \text{ C mol}^{-1} \times 0.03 \text{ V}$$

$$= -2894.61 \text{ J mol}^{-1}$$

$$= -2.89 \text{ kJ mol}^{-1}$$

Conclusion

$$\text{Again, } \Delta_{\text{r}} G^\ominus = -2.303 R T \ln K$$

$$\Rightarrow \log K = - \frac{\Delta_{\text{r}} G}{2.303 R T}$$

$$= \frac{-2894.61}{2.303 \times 8.314 \times 298}$$

$$= 0.5073$$

$$\therefore K = \text{antilog}(0.5073)$$

$$= 3.2 \text{ (approximately)}$$

$$\text{(i) } E_{\text{cell}}^\ominus = +0.34 \text{ V, } \Delta_{\text{r}} G^\ominus = -196.83 \text{ kJ mol}^{-1}, K = 3.13 \times 10^{34}; \text{ (ii)}$$

$$E_{\text{cell}}^\ominus = 0.03 \text{ V, } \Delta_{\text{r}} G^\ominus = -2.89 \text{ kJ mol}^{-1}, K = 3.2 \text{ (approximately)}$$

//

Q

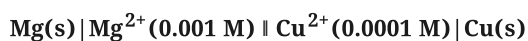
5

Nernst
equation and
emf of cells

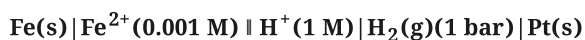


Write the Nernst equation and emf of the following cells at 298 K:

Part (i)



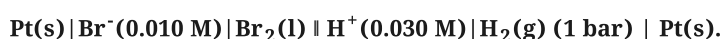
Part (ii)



Part (iii)



Part (iv)



Substitute

(i) For the given reaction, the Nernst equation can be given as:

$$E_{\text{cell}} = E_{\text{cell}}^\ominus - \frac{0.0591}{n} \log \frac{[\text{Mg}^{2+}]}{[\text{Cu}^{2+}]}$$

$$= 0.34 - (-2.36) - \frac{0.0591}{2} \log \frac{.001}{.0001}$$

$$\begin{aligned}
 &= 2.7 - \frac{0.0591}{2} \log 10 \\
 &= 2.7 - 0.02955 \\
 &= 2.67 \text{ V (approximately)}
 \end{aligned}$$

(ii) For the given reaction, the Nernst equation can be given as:

$$\begin{aligned}
 E_{\text{cell}} &= E_{\text{cell}}^{\ominus} - \frac{0.0591}{n} \log \frac{[\text{Fe}^{2+}]}{[\text{H}^+]^2} \\
 &= 0 - (-0.44) - \frac{0.0591}{2} \log \frac{0.001}{1^2} \\
 &= 0.44 - 0.02955(-3) \\
 &= 0.52865 \text{ V} \\
 &= 0.53 \text{ V (approximately)}
 \end{aligned}$$

(iii) For the given reaction, the Nernst equation can be given as:

$$\begin{aligned}
 E_{\text{cell}} &= E_{\text{cell}}^{\ominus} - \frac{0.0591}{n} \log \frac{[\text{Sn}^{2+}]}{[\text{H}^+]^2} \\
 &= 0 - (-0.14) - \frac{0.0591}{2} \log \frac{0.050}{(0.020)^2} \\
 &= 0.14 - 0.0295 \times \log 125 \\
 &= 0.14 - 0.062 \\
 &= 0.078 \text{ V} \\
 &= 0.08 \text{ V (approximately)}
 \end{aligned}$$

(iv) For the given reaction, the Nernst equation can be given as:

$$\begin{aligned}
 E_{\text{cell}} &= E_{\text{cell}}^{\ominus} - \frac{0.0591}{n} \log \frac{1}{[\text{Br}^-]^2 [\text{H}^+]^2} \\
 &= (0 - 1.09) - \frac{0.0591}{2} \log \frac{1}{(0.010)^2 (0.030)^2} \\
 &= -1.09 - 0.02955 \times \log \frac{1}{0.00000009} \\
 &= -1.09 - 0.02955 \times \log \frac{1}{9 \times 10^{-8}} \\
 &= -1.09 - 0.02955 \times \log (1.11 \times 10^7) \\
 &= -1.09 - 0.02955(0.0453 + 7) \\
 &= -1.09 - 0.208 \\
 &= -1.298 \text{ V}
 \end{aligned}$$

(i) 2.67 V (approximately); (ii) 0.53 V (approximately); (iii) 0.08 V (approximately); (iv) -1.298 V

///

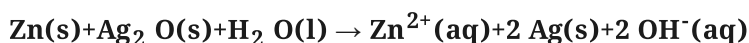
Q

6

Gibbs energy
and standard
potential of
button cell

()

In the button cells widely used in watches and other devices the following reaction takes place:



Determine $\Delta_r G^\circ$ and E° for the reaction.

Given

$$= 1.104 \text{ V}$$

Substitute

We know that,

$$\Delta_r G^\ominus = -n F E^\ominus$$

$$= -2 \times 96487 \times 1.04$$

$$= -213043.296 \text{ J}$$

$$= -213.04 \text{ kJ}$$

$$E^\ominus = 1.104 \text{ V}; \Delta_r G^\ominus = -213.04 \text{ kJ}$$

///

Q

7

Conductivity
and molar
conductivity
definitions



Define conductivity and molar conductivity for the solution of an electrolyte. Discuss their variation with concentration.

Given

Conductivity of a solution is the conductance of the solution kept between two electrodes 1 cm apart, each with an area of cross-section of 1 sq. cm. It is the inverse of resistivity, so it is also called specific conductance, and is written as κ . If ρ is the resistivity, we can write:

$$\kappa = \frac{1}{\rho}$$

At any given concentration, the conductivity of a solution is the conductance (G) of one unit volume of the solution. This volume is held between two platinum electrodes, each of unit area of cross-section, separated by unit length.

i.e.,

$$G = \kappa \frac{a}{l} = \kappa \cdot 1 = \kappa$$

(Since $a = 1, l = 1$)

Conductivity always decreases as concentration decreases, for both weak and strong electrolytes.

This is because fewer ions per unit volume are available to carry the current as concentration falls.

Molar conductivity:

At a given concentration, molar conductivity is the conductance of a volume V of solution that contains 1 mole of the electrolyte, kept between two electrodes with area of cross-section A and unit length between them.

$$\Lambda_m = \kappa \frac{A}{l}$$

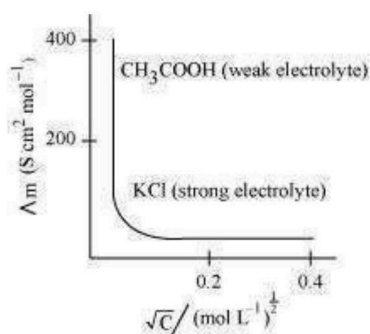
Now, $l = 1$ and $A = V$ (volume containing 1 mole of the electrolyte).

$$\therefore \Lambda_m = \kappa V$$

Molar conductivity increases as concentration decreases.

This happens because the total volume V of the solution containing one mole of the electrolyte increases on dilution.

The variation of Λ_m with $\sqrt{c}$ for strong and weak electrolytes is shown in the following plot:



The variation of Λ_m with $\sqrt{c}$ for strong and weak electrolytes.

Q

8

Molar conductivity of KCl solution



The conductivity of 0.20 M solution of KCl at 298 K is 0.0248 S cm^{-1} . Calculate its molar conductivity.

Given

Given, κ

$$= 0.0248 \text{ S cm}^{-1}$$

$$= 0.20 \text{ M}$$

Substitute

$$\therefore \text{Molar conductivity, } \Lambda_m = \frac{\kappa \times 1000}{c}$$

$$= \frac{0.0248 \times 1000}{0.2}$$

$$= 124 \text{ S cm}^2 \text{ mol}^{-1}$$

$$124 \text{ S cm}^2 \text{ mol}^{-1} //$$

Q

9

Cell constant from KCl conductivity



The resistance of a conductivity cell containing 0.001M KCl solution at 298 K is 1500Ω . What is the cell constant if conductivity of 0.001 M KCl solution at 298 K is $0.146 \times 10^{-3} \text{ S cm}^{-1}$.

Given

Given,

$$\text{Conductivity, } \kappa = 0.146 \times 10^{-3} \text{ S cm}^{-1}$$

$$\text{Resistance, } R = 1500 \Omega$$

$$\therefore \text{Cell constant} = \kappa \times R$$

$$= 0.146 \times 10^{-3} \times 1500$$

$$= 0.219 \text{ cm}^{-1}$$

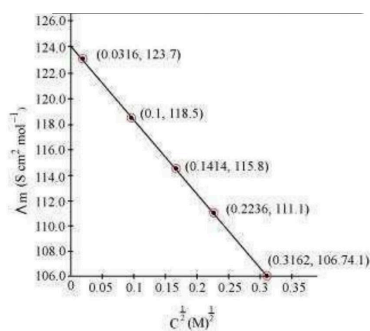
$$0.219 \text{ cm}^{-1} //$$

Q

10

Molar
conductivity
of NaCl vs
concentration

()



Plot of Λ_m against $c^{1/2}$ for the given NaCl concentration data.

The conductivity of sodium chloride at 298 K has been determined at different concentrations and the results are given below:

Concentration / M	0.001	0.010	0.020	0.050	0.100
$10^2 \times \kappa / \text{S m}^{-1}$	1.237	11.85	23.15	55.53	106.74

Calculate Λ_m for all concentrations and draw a plot between Λ_m and $c^{1/2}$. Find the value of Λ_m^0 .

Given

Given,

$$\kappa = 1.237 \times 10^{-2} \text{ S m}^{-1}, c = 0.001 \text{ M}$$

Substitute

$$\text{Then, } \kappa = 1.237 \times 10^{-4} \text{ S cm}^{-1}, c^{1/2} = 0.0316 \text{ M}^{1/2}$$

$$\therefore \Lambda_m = \frac{\kappa}{c}$$

$$= \frac{1.237 \times 10^{-4} \text{ S cm}^{-1}}{0.001 \text{ mol L}^{-1}} \times \frac{1000 \text{ cm}^3}{\text{L}}$$

$$= 123.7 \text{ S cm}^2 \text{ mol}^{-1} \text{ Given,}$$

$$\kappa = 11.85 \times 10^{-2} \text{ S m}^{-1}, c = 0.010 \text{ M}$$

$$\text{Then, } \kappa = 11.85 \times 10^{-4} \text{ S cm}^{-1}, c^{1/2} = 0.1 \text{ M}^{1/2}$$

$$\therefore \Lambda_m = \frac{\kappa}{c}$$

$$= \frac{11.85 \times 10^{-4} \text{ S cm}^{-1}}{0.010 \text{ mol L}^{-1}} \times \frac{1000 \text{ cm}^3}{\text{L}}$$

$$= 118.5 \text{ S cm}^2 \text{ mol}^{-1} \text{ Given,}$$

$$\kappa = 23.15 \times 10^{-2} \text{ S m}^{-1}, c = 0.020 \text{ M}$$

$$\text{Then, } \kappa = 23.15 \times 10^{-4} \text{ S cm}^{-1}, c^{1/2} = 0.1414 \text{ M}^{1/2}$$

$$\therefore \Lambda_m = \frac{\kappa}{c}$$

$$= \frac{23.15 \times 10^{-4} \text{ S cm}^{-1}}{0.020 \text{ mol L}^{-1}} \times \frac{1000 \text{ cm}^3}{\text{L}}$$

$$= 115.8 \text{ S cm}^2 \text{ mol}^{-1} \text{ Given,}$$

$$\kappa = 55.53 \times 10^{-2} \text{ S cm}^{-1}, c = 0.050 \text{ M}$$

$$\text{Then, } \kappa = 55.53 \times 10^{-4} \text{ S cm}^{-1}, c^{1/2} = 0.2236 \text{ M}^{1/2}$$

$$\therefore \Lambda_m = \frac{\kappa}{c}$$

$$= \frac{55.53 \times 10^{-4} \text{ S cm}^{-1}}{0.050 \text{ mol L}^{-1}} \times \frac{1000 \text{ cm}^3}{\text{L}}$$

$$= 111.11 \text{ S cm}^2 \text{ mol}^{-1} \text{ Given,}$$

$$\kappa = 106.74 \times 10^{-2} \text{ S cm}^{-1}, c = 0.100 \text{ M}$$

$$\text{Then, } \kappa = 106.74 \times 10^{-4} \text{ S cm}^{-1}, c^{1/2} = 0.3162 \text{ M}^{1/2}$$

$$\Lambda_m = \frac{\kappa}{c}$$

$$= \frac{106.74 \times 10^{-4} \text{ S cm}^{-1}}{0.100 \text{ mol L}^{-1}} \times \frac{1000 \text{ cm}^3}{\text{L}}$$

$$= 106.74 \text{ S cm}^2 \text{ mol}^{-1} \text{ Now, we}$$

have the following data:

$c^{1/2} / \text{M}^{1/2}$	0.0316	0.1	0.1414	0.2236	0.3162
$\Lambda_m (\text{S cm}^2 \text{ mol}^{-1})$	123.7	118.5	115.8	111.1	106.74

Conclusion

Since the line intercepts Λ_m at $124.0 \text{ S cm}^2 \text{ mol}^{-1}$, $\Lambda_m^0 = 124.0 \text{ S cm}^2 \text{ mol}^{-1}$.

$$\Lambda_m^0 = 124.0 \text{ S cm}^2 \text{ mol}^{-1} //$$

Q

11

Molar conductivity and dissociation constant of acetic acid



Conductivity of 0.00241 M acetic acid is $7.896 \times 10^{-5} \text{ S cm}^{-1}$. Calculate its molar conductivity. If Λ_m^0 for acetic acid is $390.5 \text{ S cm}^2 \text{ mol}^{-1}$, what is its dissociation constant?

Given

$$\begin{aligned} \text{Given, } \kappa &= 7.896 \times 10^{-5} \text{ S cm}^{-1} \\ &= 0.00241 \text{ mol L}^{-1} \end{aligned}$$

Substitute

Then, molar conductivity,

Key formula

$$\Lambda_m = \frac{\kappa}{c}$$



Substitute

$$= \frac{7.896 \times 10^{-5} \text{ S cm}^{-1}}{0.00241 \text{ mol L}^{-1}} \times \frac{1000 \text{ cm}^3}{\text{L}}$$

$$= 32.76 \text{ S cm}^2 \text{ mol}^{-1}$$

$$\Lambda_m^0 = 390.5 \text{ S cm}^2 \text{ mol}^{-1} \text{ Again,}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^0} = \frac{32.76 \text{ S cm}^2 \text{ mol}^{-1}}{390.5 \text{ S cm}^2 \text{ mol}^{-1}}$$

Now,

$$= 0.084$$

$$\therefore K_a = \frac{c \alpha^2}{(1-\alpha)}$$

$$= \frac{(0.00241 \text{ mol L}^{-1})(0.084)^2}{(1-0.084)}$$

$$= 1.86 \times 10^{-5} \text{ mol L}^{-1}$$

$$\Lambda_m = 32.76 \text{ S cm}^2 \text{ mol}^{-1}; \alpha = 0.084; K_a = 1.86 \times 10^{-5} \text{ mol L}^{-1} //$$

Q

12

Charge
required for
reductions



How much charge is required for the following reductions:

Part (i)

1 mol of Al^{3+} to Al ?

Part (ii)

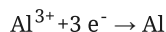
1 mol of Cu^{2+} to Cu ?

Part (iii)

1 mol of MnO_4^- to Mn^{2+} ?

Key formula

(i)



Substitute

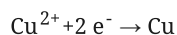
$\therefore$ Required charge = 3 F

$$= 3 \times 96487 \text{ C}$$

$$= 289461 \text{ C}$$

Key formula

(ii)



Substitute

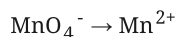
$\therefore$ Required charge = 2 F

$$= 2 \times 96487 \text{ C}$$

$$= 192974 \text{ C}$$

Key formula

(iii)



Substitute

i.e.,

$\therefore$

$$= 5 \times 96487 \text{ C}$$

$$= 482435 \text{ C}$$

(i) 289461 C (3 F); (ii) 192974 C (2 F); (iii) 482435 C (5 F) //

Q

13

Faraday
charge for
metal
production

How much electricity in terms of Faraday is required to produce

Part (i)

20.0 g of Ca from molten CaCl_2 ?

Part (ii)

40.0 g of Al from molten Al_2O_3 ?

Key formula

(i) According to the question,



(40 g)

Substitute

Electricity required to produce 40 g of calcium = 2 F

$$\text{Therefore, electricity required to produce 20 g of calcium} = \frac{2 \times 20}{40} \text{ F}$$

$$= 1 \text{ F}$$

Key formula

(ii) According to the question,



(27 g)

Substitute

Electricity required to produce 27 g of Al = 3 F

$$\text{Therefore, electricity required to produce 40 g of Al} = \frac{3 \times 40}{27} \text{ F} = 4.44 \text{ F}$$

(i) 1 F; (ii) 4.44 F //

Q

14

Charge for
oxidation
reactions

()

How much electricity is required in coulomb for the oxidation of

Part (i)

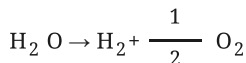
1 mol of H_2O to O_2 ?

Part (ii)

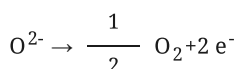
1 mol of FeO to Fe_2O_3 ?

Key formula

(i) According to the question,



Now, we can write:



Substitute

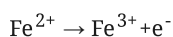
Electricity required for the oxidation of 1 mol of H_2O to O_2 = 2 F

$$= 2 \times 96487 \text{ C}$$

$$= 192974 \text{ C}$$

Key formula

(ii) According to the question,



Substitute

Electricity required for the oxidation of 1 mol of FeO to Fe_2O_3 = 1 F

$$= 96487 \text{ C}$$

(i) 192974 C (2 F); (ii) 96487 C (1 F) //

Q

15

Mass of
nickel
deposited

A solution of $\text{Ni}(\text{NO}_3)_2$ is electrolysed between platinum electrodes using a current of 5 amperes for 20 minutes. What mass of Ni is deposited at the cathode?

Given

Given,

Current = 5 A

Time = $20 \times 60 = 1200$ s $\therefore$ Charge = current $\times$ time= 5×1200

= 6000 C

Key formula

According to the reaction,



58.7 g

Substitute

Nickel deposited by 2×96487 C = 58.71 g

Therefore, nickel deposited by 6000 C = $\frac{58.71 \times 6000}{2 \times 96487}$ g

= 1.825 g

Conclusion

Hence, 1.825 g of nickel will be deposited at the cathode.

1.825 g of nickel

///

Q

16

Series
electrolytic
cells
deposition

Three electrolytic cells A,B,C containing solutions of ZnSO_4 , AgNO_3 and CuSO_4 , respectively are connected in series. A steady current of 1.5 amperes was passed through them until 1.45 g of silver deposited at the cathode of cell B. How long did the current flow? What mass of copper and zinc were deposited?

Key formula

According to the reaction:



108 g

Substitute

i.e., 108 g of Ag is deposited by 96487 C.

Therefore, 1.45 g of Ag is deposited by = $\frac{96487 \times 1.45}{108}$ C

= 1295.43 C

Given,

Current = 1.5 A

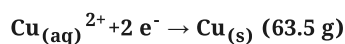
$\therefore$ Time = $\frac{1295.43}{1.5}$ s

= 863.6 s

= 864 s

= 14.40 min

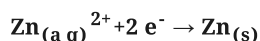
Again,



i.e., $2 \times 96487 \text{ C of charge deposit} = 63.5 \text{ g of Cu}$

$$\text{Therefore, } 1295.43 \text{ C of charge will deposit} = \frac{63.5 \times 1295.43}{2 \times 96487} \text{ g}$$

$$= 0.426 \text{ g of Cu}$$



$$65.4 \text{ g}$$

i.e., $2 \times 96487 \text{ C of charge deposit} = 65.4 \text{ g of Zn}$

$$\text{Therefore, } 1295.43 \text{ C of charge will deposit} = \frac{65.4 \times 1295.43}{2 \times 96487} \text{ g}$$

$$= 0.439 \text{ g of Zn}$$

Time = 864 s (14.40 min); Cu deposited = 0.426 g; Zn deposited = 0.439 g

//

Q

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Feasibility of
redox
reactions



Using the standard electrode potentials given in Table 3.1, predict if the reaction between the following is feasible:

Part (i)

$\text{Fe}^{3+}(\text{aq})$ and $\text{I}^{-}(\text{aq})$

Part (ii)

$\text{Ag}^{+}(\text{aq})$ and $\text{Cu}(\text{s})$

Part (iii)

$\text{Fe}^{3+}(\text{aq})$ and $\text{Br}^{-}(\text{aq})$

Part (iv)

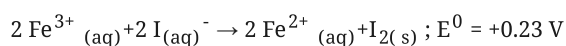
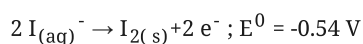
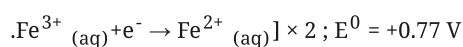
$\text{Ag}(\text{s})$ and $\text{Fe}^{3+}(\text{aq})$

Part (v)

$\text{Br}_2(\text{aq})$ and $\text{Fe}^{2+}(\text{aq})$.

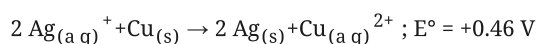
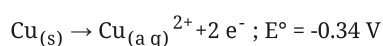
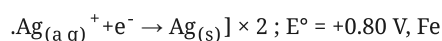
Conclusion

(i)



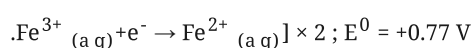
Since $\text{Fe}^{3+}(\text{aq})$ and $\text{I}^{-}(\text{aq})$ is feasible.

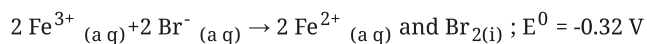
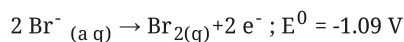
(ii)



Since E^0 for the overall reaction is positive, the reaction between $\text{Ag}^{+}(\text{aq})$ and $\text{Cu}_{(\text{s})}$ is feasible.

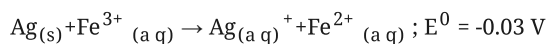
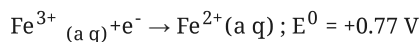
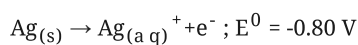
(iii)



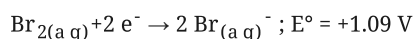


Since E^0 for the overall reaction is negative, the reaction between $\text{Fe}^{3+} (\text{aq})$ and $\text{Br}^- (\text{aq})$ is not feasible.

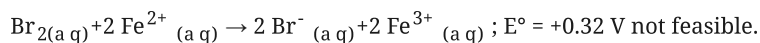
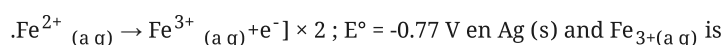
(iv)



(iv)



!



Since E^0 for the overall reaction is positive, the reaction between $\text{Br}_{2(\text{aq})}$ and $\text{Fe}^{2+} (\text{aq})$ is feasible.

Q

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Products of electrolysis

(())

Predict the products of electrolysis in each of the following:

Part (i)

An aqueous solution of AgNO_3 with silver electrodes.

Part (ii)

An aqueous solution of AgNO_3 with platinum electrodes.

Part (iii)

A dilute solution of H_2SO_4 with platinum electrodes.

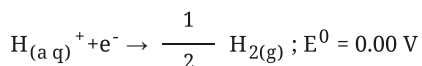
Part (iv)

An aqueous solution of CuCl_2 with platinum electrodes.

Given

(i) At cathode:

The following reduction reactions compete to take place at the cathode.



Conclusion

The reaction with a higher value of E^0 takes place at the cathode. Therefore, deposition of silver will take place at the cathode.

Substitute

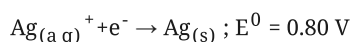
At anode:

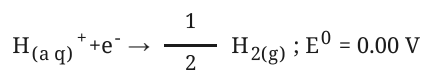
The Ag anode is attacked by NO_3^- ions. Therefore, the silver electrode at the anode dissolves in the solution to form Ag^+ .

(ii) At cathode:

Given

The following reduction reactions compete to take place at the cathode.



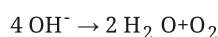
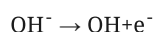
**Conclusion**

The reaction with a higher value of E° takes place at the cathode. Therefore, deposition of silver will take place at the cathode.

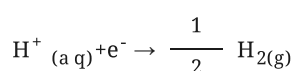
Substitute

At anode:

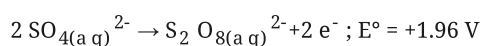
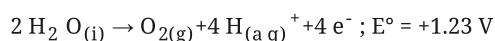
Since Pt electrodes are inert, the anode is not attacked by NO_3^{-} ions. Therefore, OH^{-} or NO_3^{-} ions can be oxidized at the anode. But OH^{-} ions have a lower discharge potential, so they get preference and decompose to liberate O_2 .



(iii) At the cathode, the following reduction reaction occurs to produce H_2 gas.

Given

At the anode, the following processes are possible.

**Conclusion**

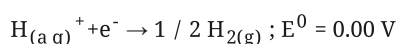
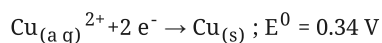
For dilute sulphuric acid, reaction (i) is preferred to produce O_2 gas. But for concentrated sulphuric acid, reaction (ii) occurs.

Substitute

(iv) At cathode:

Given

The following reduction reactions compete to take place at the cathode.

**Conclusion**

The reaction with a higher value of E° takes place at the cathode. Therefore, deposition of copper will take place at the cathode.

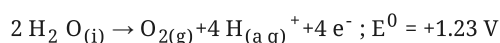
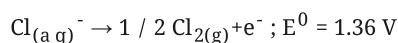
Substitute

At anode:

Given

The following oxidation reactions are possible at the anode.

At the anode, the reaction with a lower value of

**Conclusion**

E° is preferred. But due to the overpotential of oxygen, Cl^{-} gets oxidized at the anode to produce Cl_2 gas.